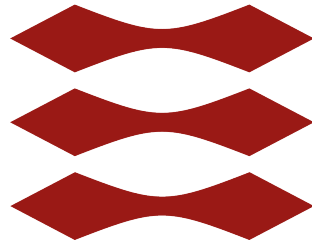


DTU



TECHNICAL UNIVERSITY OF DENMARK

25314 - COMPUTATIONAL MARINE ECOLOGICAL MODELLING

Report 1: NPZDO model

Created by:
Manon Cotoni s253043

DTU - Kongens Lyngby (Denmark)
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1 Illustration

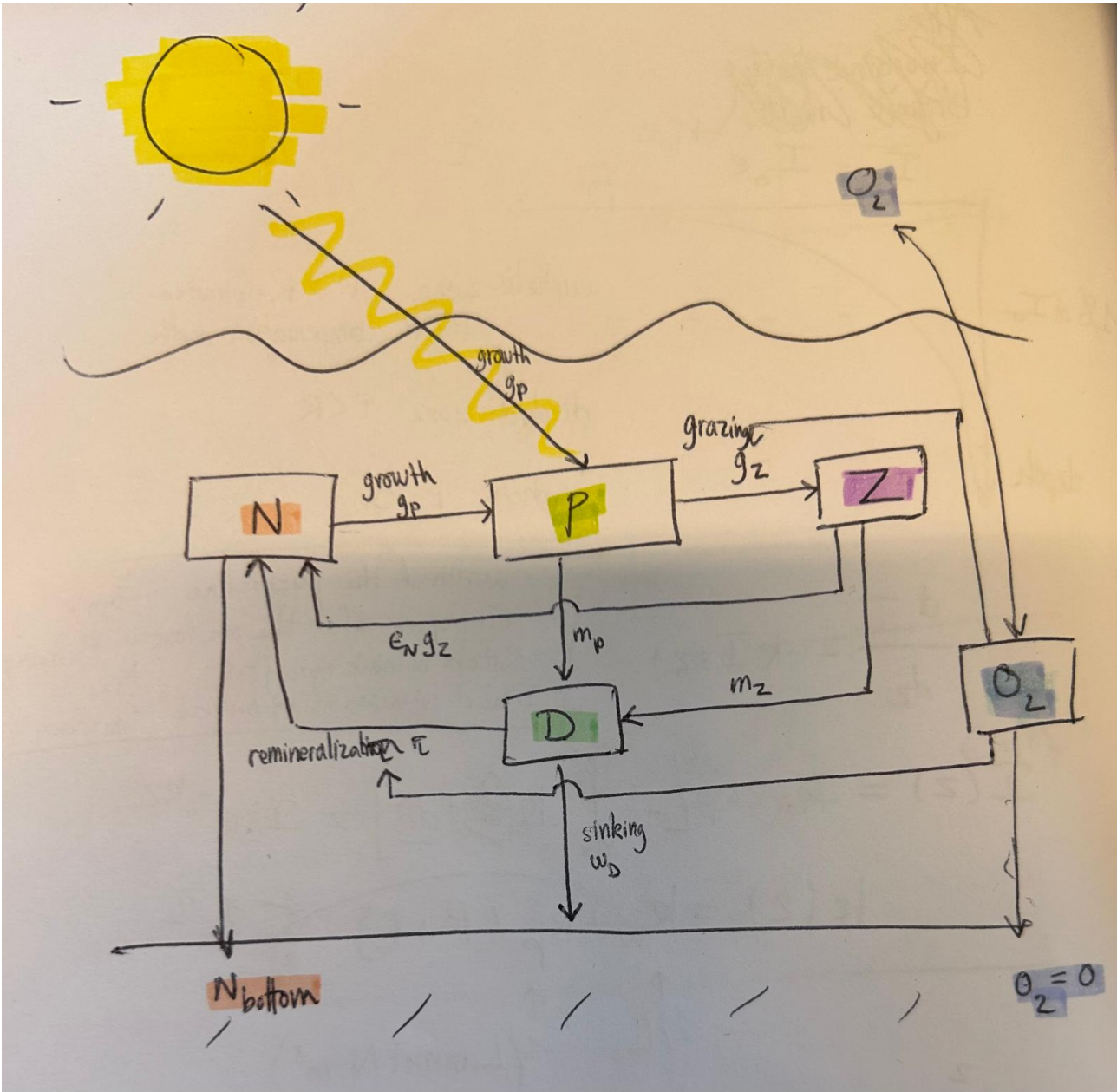


Figure 1: NPZDO model illustration

2 Governing equations

The NPZDO model (Nutrient, Phytoplankton, Zooplankton, Detritus, Oxygen) is governed by a system of one-dimensional advection-diffusion-reaction equations. For a tracer concentration $C(z, t)$, the general conservation law is:

$$\frac{\partial C}{\partial t} = \text{Reactions} + \frac{\partial}{\partial z} \left(\kappa(z, t) \frac{\partial C}{\partial z} \right) - \frac{\partial(wC)}{\partial z} \quad (1)$$

The specific system for the five state variables is defined as:

$$\frac{\partial N}{\partial t} = -g_P P + \epsilon_N g_Z Z + \tau(O) D + \frac{\partial}{\partial z} \left(\kappa \frac{\partial N}{\partial z} \right) \quad (2)$$

$$\frac{\partial P}{\partial t} = g_P P - g_Z Z - m_P P + \frac{\partial}{\partial z} \left(\kappa \frac{\partial P}{\partial z} \right) - w_P \frac{\partial P}{\partial z} \quad (3)$$

$$\frac{\partial Z}{\partial t} = (1 - \epsilon_N - \epsilon_D) g_Z Z - m_Z Z^2 + \frac{\partial}{\partial z} \left(\kappa \frac{\partial Z}{\partial z} \right) \quad (4)$$

$$\frac{\partial D}{\partial t} = m_P P + m_Z Z + \epsilon_D g_Z Z - \tau(O) D + \frac{\partial}{\partial z} \left(\kappa \frac{\partial D}{\partial z} \right) - w_D \frac{\partial D}{\partial z} \quad (5)$$

$$\frac{\partial O}{\partial t} = y_P g_P P - y_N \epsilon_N g_Z Z - y_N \tau(O) D + \frac{\partial}{\partial z} \left(\kappa \frac{\partial O}{\partial z} \right) \quad (6)$$

2.1 Phytoplankton Growth (g_P)

Growth is regulated by light I and nutrients N :

$$g_P = g_{P,max} \cdot \min \left(\frac{I}{I + k_L}, \frac{N}{N + k_N} \right) \quad (7)$$

2.2 Grazing and Remineralization

Zooplankton consumption g_Z and remineralization rate τ are both oxygen-dependent via a sigmoid function $\sigma(O)$:

$$g_Z(O) = g_{Z,max} \frac{P}{P + k_Z} \sigma(O), \quad \tau(O) = r \sigma(O), \quad \sigma(O) = \frac{1}{1 + e^{-O}} \quad (8)$$

2.3 Seasonal light variation

The light available at depth z is a function of the seasonal surface irradiance $I_0(t)$ and vertical attenuation:

$$I(z, t) = I_0(t) \cdot \exp \left(-k_W z - k_P \int_0^z P(z') dz' \right) \quad (9)$$

To simulate annual cycles, the surface light $I_0(t)$ follows a sinusoidal seasonal variation:

$$I_0(t) = 0.5 \left[1 + \sin \left(\frac{2\pi t}{365} \right) \right] (I_{0,summer} - I_{0,winter}) + I_{0,winter} \quad (10)$$

where $I_{0,summer}$ is the summer peak surface light and $I_{0,winter}$ is the minimum winter irradiance. Light attenuates with depth based on the background turbidity of seawater (k_W) and the self-shading effect of the phytoplankton biomass (k_P).

2.4 Seasonal diffusivity

The vertical mixing coefficient $\kappa(z, t)$ is modeled to represent a seasonally varying mixed-layer depth. The diffusivity profile follows a hyperbolic tangent transition from high surface mixing (κ_0) to low deep-water mixing (κ_{min}):

$$\kappa(z, t) = \kappa_{min} + \frac{\kappa_0 - \kappa_{min}}{2} \left[1 - \tanh \left(\frac{z - z_0(t)}{\zeta} \right) \right] \quad (11)$$

where ζ is the transition thickness and $z_0(t)$ is the time-dependent depth of the pycnocline. To simulate the deepening of the mixed layer during winter and its shoaling in spring, the pycnocline depth is forced by:

$$z_0(t) = \frac{1}{2} \left(1 - \sin \left[\frac{2\pi(t - t_{MaxSpring})}{365} \right] \right)^{\zeta_{MaxSteep}} (z_{MixWinter} - z_{MixSummer}) + z_{MixSummer} \quad (12)$$

where $t_{MaxSpring}$ represents the day of the year when the spring pycnocline begins to form; $z_{MixWinter}$ and $z_{MixSummer}$ are the maximum and minimum mixed-layer depths, respectively; and $\zeta_{MaxSteep}$ is the steepness exponent, which controls the duration of the stratified period versus the deeply mixed period.

3 Parameters

Table 1: Model Parameters for NPZD-Oxygen Dynamics

Parameter Description	Symbol	Value	Unit
<i>Water Column</i>			
Total depth	H	20.0	m
Number of vertical cells	n	100	-
Upper-ocean diffusivity	κ_0	10.0	m ² /day
Deep diffusivity	κ_{min}	5.0	m ² /day
Thermocline depth (mixed layer depth)	$z_0, z_{MixSummer}$	10.0	m
Transition thickness	ζ	2.0	m
Bottom nutrient concentration	N_{bottom}	10.0	mmol/m ³
<i>Light & Seasonality</i>			
Default surface light intensity (in summer)	$I_0, I_{0,summer}$	1400.0	$\mu\text{mol ph/m}^2/\text{s}$
Winter min. surface light	$I_{0,winter}$	0.0	$\mu\text{mol ph/m}^2/\text{s}$
Light atten. (water)	k_W	0.15	m ⁻¹
Light atten. (plankton)	k_P	0.05	m ² /mmol N
Day of pycnocline formation	$t_{MaxSpring}$	75.0	day
Day of max light	$t_{LightMax}$	174.0	day
Winter mixed-layer depth	$z_{MixWinter}$	18.0	m
Pycnocline slope exponent	$\zeta_{MaxSteep}$	4.0	-
<i>Phytoplankton</i>			
Max growth rate	$g_{P,max}$	1.5	day ⁻¹
Light half-saturation	k_L	50.0	$\mu\text{mol ph/m}^2/\text{s}$
Nutrient half-saturation	k_N	0.4	mmol/m ³
Mortality rate	m_P	0.25	day ⁻¹
Sinking speed	w_P	0.0	m/day
<i>Zooplankton</i>			
Max grazing rate	$g_{Z,max}$	0.7	day ⁻¹
Half-sat. grazing	k_Z	3.0	mmol/m ³
Respiration fraction	ϵ_N	0.3	-
Egestion fraction	ϵ_D	0.3	-
Mortality rate	m_Z	0.2	day ⁻¹ / mmol / m ³
<i>Detritus</i>			
Remineralization rate	r	0.05	day ⁻¹
Sinking speed	w_D	5.0	m/day
<i>Oxygen</i>			
mmol O ₂ produced per mmol N assimilated	y_P	9.0	mmol O ₂ /mmol N
mmol O ₂ consumed per mmol N remineralized	y_N	6.625	mmol O ₂ /mmol N
Surface gas exchange coeff.	$K_{O_2,surf}$	20.0	m/day
Bottom gas exchange coeff.	$K_{O_2,bot}$	1.0	m/day
Atmospheric O ₂ saturation	$O_{2,atm}$	260.0	mmol O ₂ /m ³

4 Numerical scheme

4.1 Spatial Discretization

The water column of depth H is divided into n cells of thickness $\Delta z = H/n$. The concentration C_i is defined at the center of each cell:

$$z_i = \left(i + \frac{1}{2}\right) \Delta z, \quad i = 0, \dots, n-1. \quad (13)$$

Fluxes J_i are defined at the interfaces between cells:

$$z_{i+\frac{1}{2}} = i\Delta z, \quad i = 0, \dots, n. \quad (14)$$

The discrete conservation equation for each cell is:

$$\frac{dC_i}{dt} = -\frac{J_{i+1} - J_i}{\Delta z}. \quad (15)$$

4.2 Advection and diffusion

The advective flux is discretized using an upwind scheme:

$$J_{a,i} = wC_{i-1}. \quad (16)$$

The diffusive flux is discretized using a central difference scheme:

$$J_{d,i} = -\kappa_i \frac{C_i - C_{i-1}}{\Delta z}. \quad (17)$$

4.3 Boundary Conditions

Variable	Surface Flux (J_0)	Bottom Flux (J_n)
Nutrients (N)	$J_0 = 0$	$J_n = -\kappa_n \frac{N_{bottom} - N_n}{\Delta z}$
Phytoplankton (P)	$J_0 = 0$	$J_n = w_P \cdot P_n$
Zooplankton (Z)	$J_0 = 0$	$J_n = 0$
Detritus (D)	$J_0 = 0$	$J_n = w_D \cdot D_n$
Oxygen (O)	$J_0 = -K_{O_2,surf}(O_1 - O_{atm})$	$J_n = -K_{O_2,bot}(O_{bottom} - O_n)$

Table 2: Boundary flux expressions (O_{bottom} is assumed to be 0)

5 Convergence

To show that the solution is converged, we plotted the evolution of the depth distribution of each state-variable concentration over time. We observe that the solution converges (does not change anymore) rapidly within approximately 50 days.

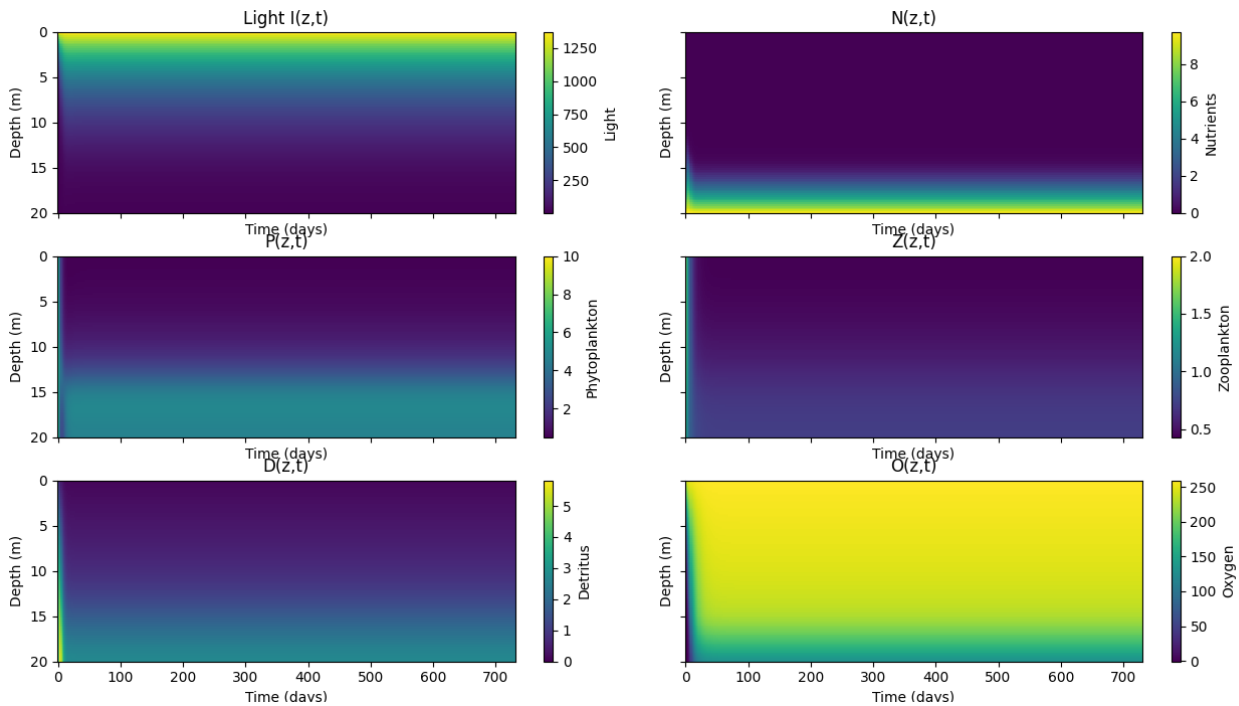


Figure 2: Convergence plot of the NPZDO model after one year ($n = 100$)

And to show that the solution is independent of the grid size, we plotted it for different values of n , the number of cells. We observe that in all four cases, the converged solution is unique, which proves the independence of the solution when it comes to the grid size. The only change the grid size brings is the resolution of the plots.

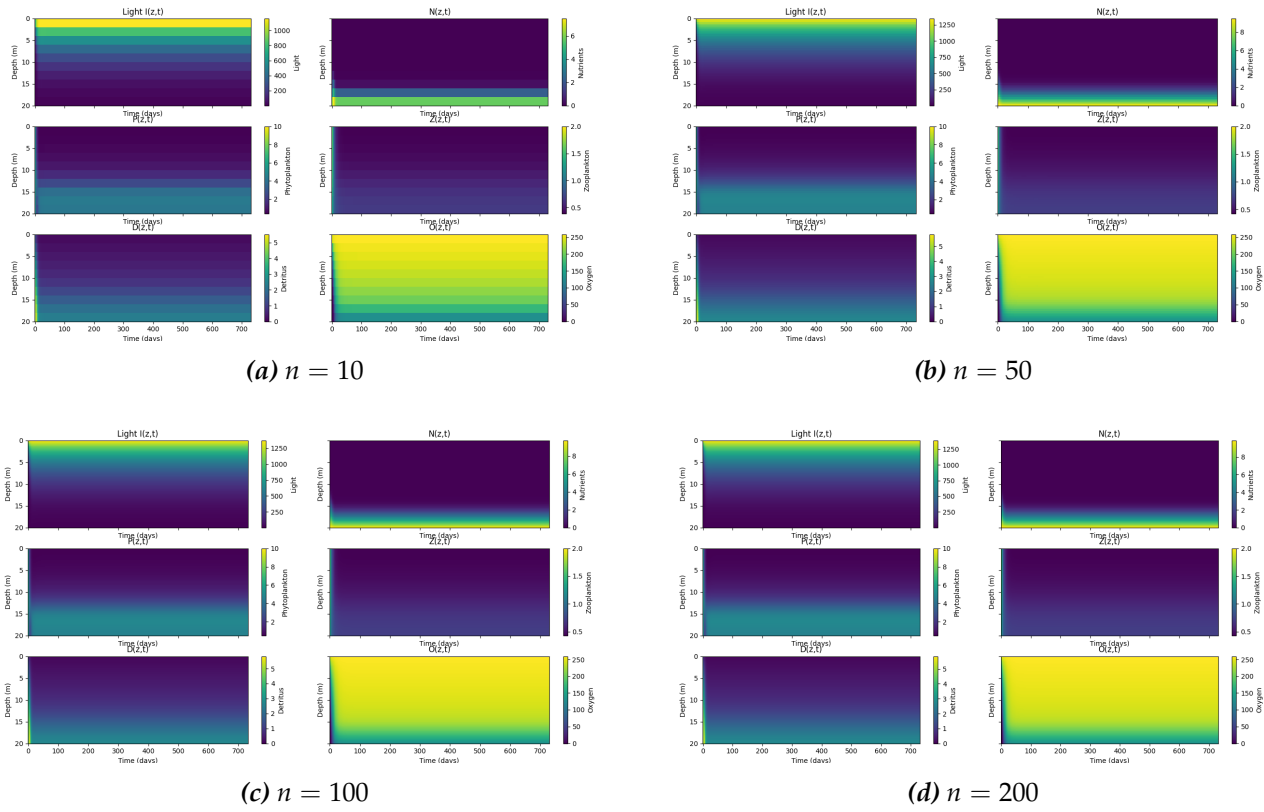


Figure 3: Convergence plots of the NPZDO model after one year for different vertical grid resolutions

NB: Here, we plotted the solution using phytoplankton self-shading in light profile (`use_phyto_light_damping` = True) but without including the seasonal light or the seasonal diffusivity (`use_seasonal_light` = False, `use_seasonal_diffusivity` = False).

6 Phytoplankton growth limiting factors

Phytoplankton growth is regulated by light I and nutrients N :

$$g_P = g_{P,max} \cdot \min \left(\frac{I}{I + k_L}, \frac{N}{N + k_N} \right) \quad (18)$$

So phytoplankton growth can either be limited by light or by nutrient.

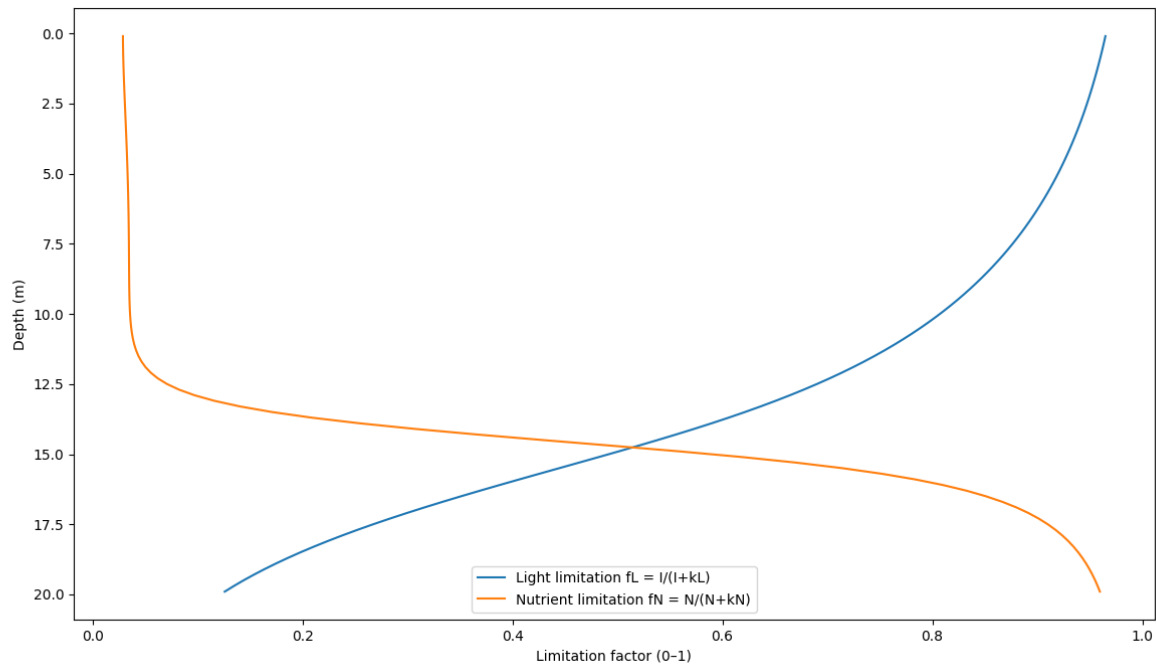


Figure 4: Phytoplankton limiting factors vs depth

We observe that, for the default parameters values listed in Table 1, nutrients are the limiting factor in the first 15 meters of the water column, while light becomes the limiting factor in the remaining 5 meters.

NB: To plot phytoplankton limiting factors, we runned the model using phytoplankton self-shading in light profile (`use_phyto_light_damping = True`) but without including the seasonal light or the seasonal diffusivity (`use_seasonal_light = False`, `use_seasonal_diffusivity = False`).

7 Summer steady-state solutions

We plotted the steady-state solutions of the model (after 5 years) under stable summer conditions, meaning using the parameters values listed in Table 1 with a surface light of 1400 $\mu\text{mol photons/m}^2/\text{s}$.

NB: To observe the stable summer conditions, we plotted the solutions using phytoplankton self-shading in light profile (`use_phyto_light_damping = True`) but without including the seasonal light or the seasonal diffusivity (`use_seasonal_light = False`, `use_seasonal_diffusivity = False`).

7.1 Steady-state solutions of the model under stable summer conditions: baseline scenario

The baseline scenario is runned with a bottom nutrient concentration of 10 mmol N/m^3 .

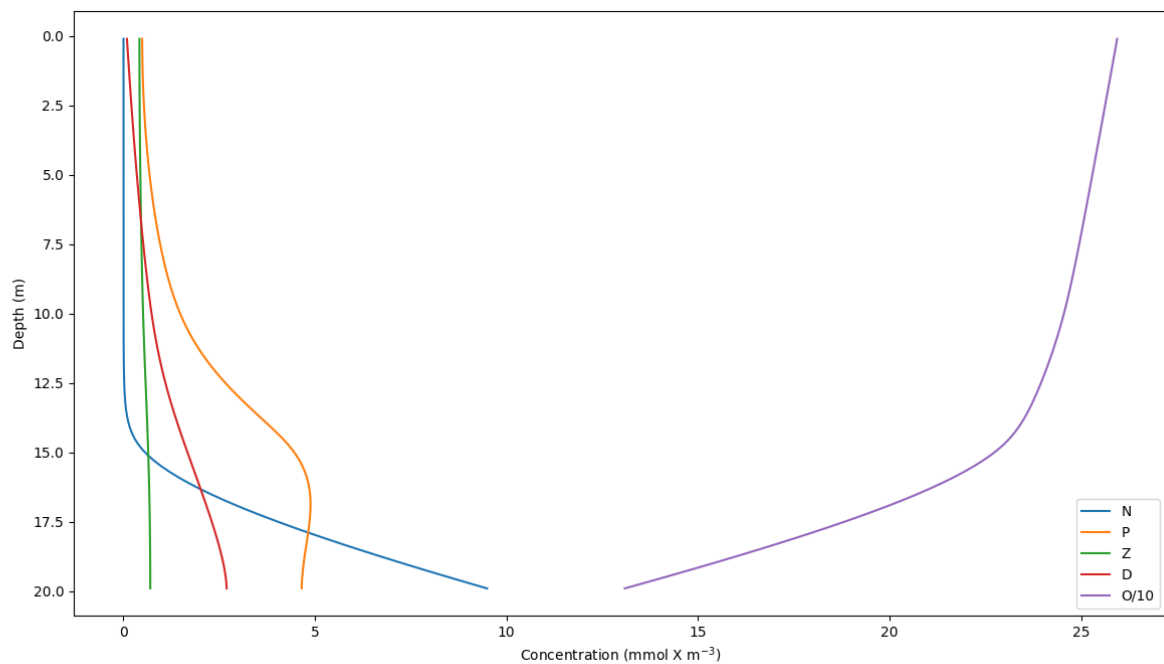


Figure 5: Final NPZDO profiles: $N_{\text{bottom}}=10 \text{ mmol N m}^{-3}$

The system exhibits characteristics of an oligotrophic summer water column. Nutrient concentrations remain very low in the upper layers due to uptake by phytoplankton and limited vertical mixing across the stratified water column. As a result, phytoplankton biomass remains relatively small and forms a maximum around mid-depth (15 meters and below) where some nutrients become available. Zooplankton concentrations are modest and broadly follow the phytoplankton distribution due to grazing dependence. Detritus concentrations remain low because primary production is limited. Oxygen concentrations are relatively high throughout the water column, reflecting moderate respi-

ration rates and limited decomposition of organic matter.

7.2 Steady-state solutions of the model under stable summer conditions: high nutrient input

To run a scenario with high nutrient input, we increased the bottom nutrient concentration to 100 mmol N/m³.

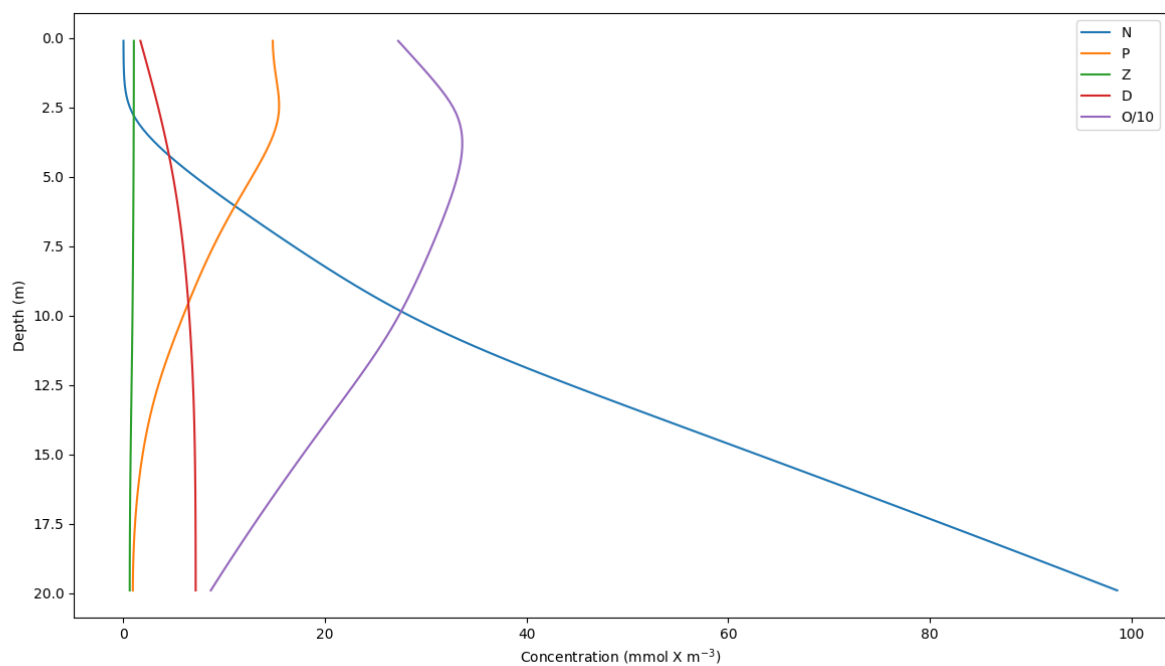


Figure 6: Final NPZDO profiles: $N_{bottom}=100 \text{ mmol N m}^{-3}$

Elevated nutrient concentrations support substantially higher phytoplankton biomass, particularly in the upper water column where light is available. This leads to a strong phytoplankton bloom near the surface (from 0 to 3 meters depth). The larger phytoplankton biomass sustains higher zooplankton abundance through grazing interactions. Increased biological production also generates larger amounts of detritus, which accumulates and sinks through the water column. As detrital organic matter is remineralized at depth, oxygen consumption increases, resulting in lower oxygen concentrations in deeper layers compared to the baseline case.

7.3 Comparison of the two situations: high vs low nutrient loading

To investigate the eutrophication level, we calculated the chlorophyll-a concentration using a N:Chl ratio of 8 gN:gChl to represent ocean and mesotrophic conditions.

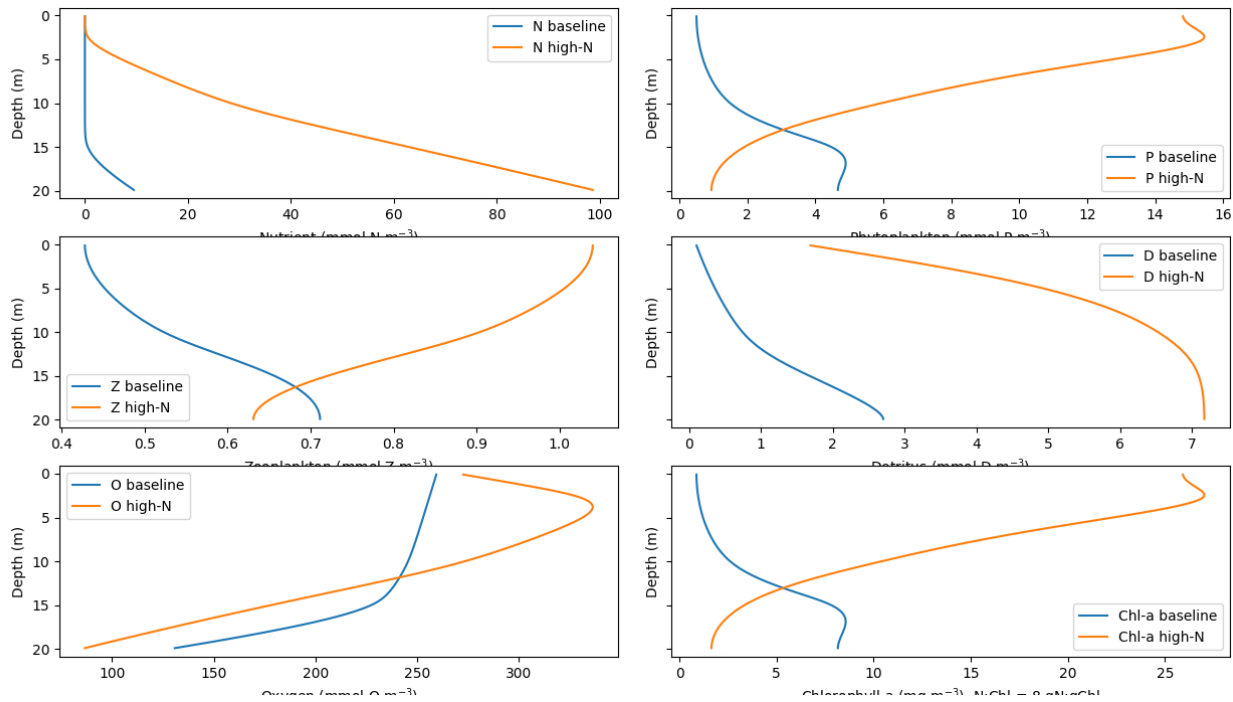
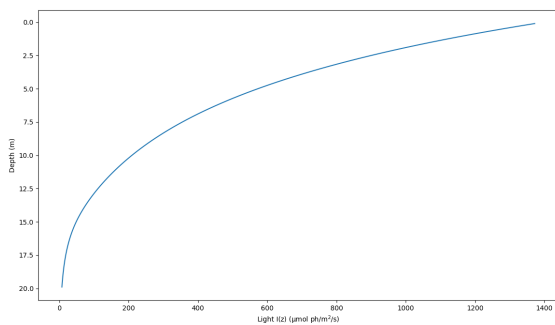
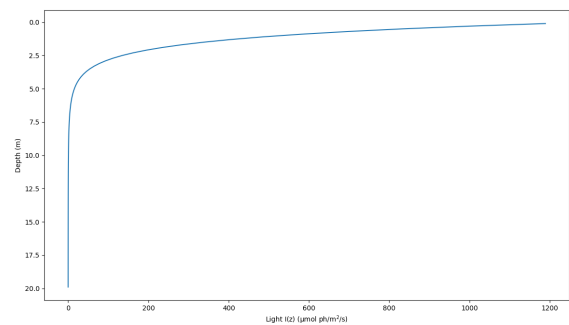


Figure 7: Baseline ($N_{\text{bottom}}=10 \text{ mmol N m}^{-3}$) vs high nutrient input ($N_{\text{bottom}}=100 \text{ mmol N m}^{-3}$)

In the low-nutrient case, phytoplankton growth is nutrient-limited, constraining the entire food web and maintaining low biomass levels. In the high-nutrient case, nutrient limitation is relaxed, allowing strong primary production that propagates through trophic levels and increases organic matter production and remineralization. This enhanced biological activity also intensifies oxygen consumption at depth. Because summer stratification limits vertical mixing, these biological processes create pronounced vertical gradients, leading to distinct ecosystem states under low versus high nutrient loading.



(a) $N_{\text{bottom}}=10 \text{ mmol N m}^{-3}$



(b) $N_{\text{bottom}}=100 \text{ mmol N m}^{-3}$

Figure 8: Final light profile: low vs high nutrient input

Under low nutrient loading, phytoplankton biomass remains small because growth is strongly nutrient-limited, so light attenuation is dominated mainly by the background absorption of water. This allows light to penetrate relatively deep into the water column, producing a gradual decline

in light with depth. In contrast, high nutrient loading stimulates strong phytoplankton growth in surface waters, and the resulting high biomass increases self-shading and light absorption. Consequently, light intensity decreases much more rapidly with depth, confining most primary production to the upper layers in eutrophic systems, whereas in oligotrophic systems photosynthesis can occur over a larger vertical range.

8 Sensitivity analysis

For this sensitivity analysis, we run the model for different values of the parameters under a standard summer condition, meaning with a surface light intensity of $1400 \mu\text{mol ph/m}^2/\text{s}$.

NB: To carry out the sensitivity analysis, we runned the model using phytoplankton self-shading in light profile (`use_phyto_light_damping = True`) but without including the seasonal light or the seasonal diffusivity (`use_seasonal_light = False`, `use_seasonal_diffusivity = False`).

8.1 Light half saturation coefficient

The default value of light half saturation coefficient is $k_L = 50.0 \mu\text{mol ph/m}^2/\text{s}$. For this sensitivity analysis, we increase the value by approximatively 25% using $k_L = 62.5 \mu\text{mol ph/m}^2/\text{s}$.

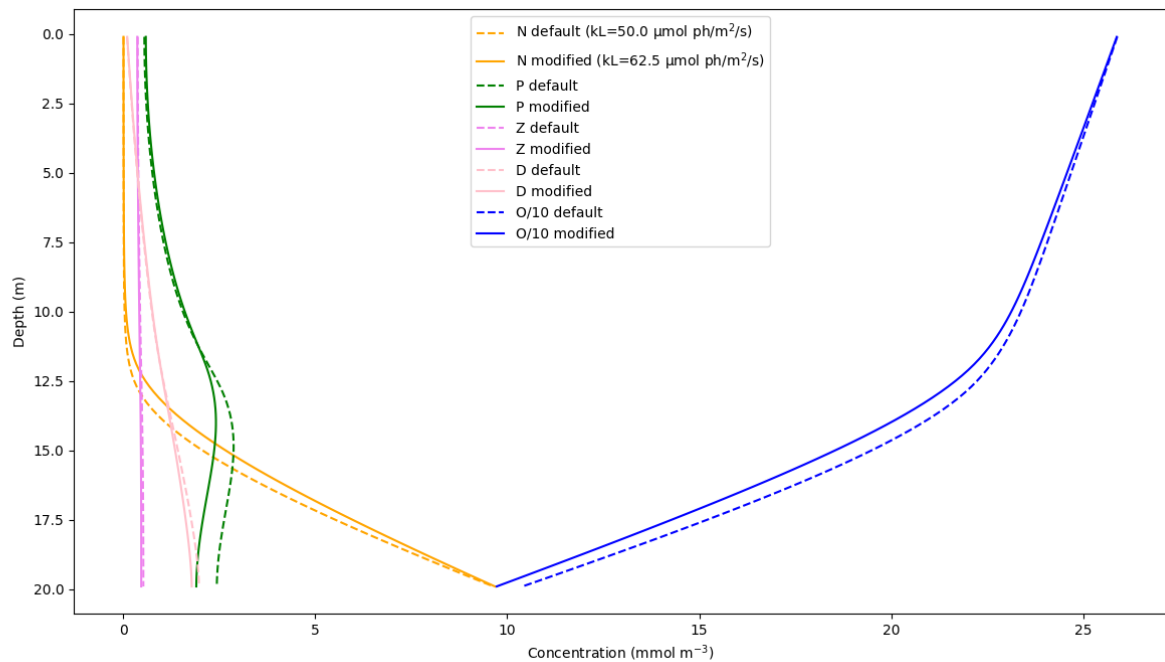


Figure 9: Final NPZDO profiles: light half saturation coefficient impact

Increasing the light half saturation coefficient causes phytoplankton population to decrease and its bump to be a bit closer to the surface compared to the baseline model. Moreover, it also causes oxygen concentration to decrease as well as detritus even though it is not striking. Zooplankton is almost unchanged (very little decrease). If all state variables appear to be decreasing, concentration of nutrients, on the other hand, increases slightly.

8.2 Phytoplankton maximum growth rate

The default value of phytoplankton maximum growth rate is $g_{P,max} = 1.5 \text{ day}^{-1}$. For this sensitivity analysis, we increase the value by approximatively 25% using $g_{P,max} = 2.0 \text{ day}^{-1}$.

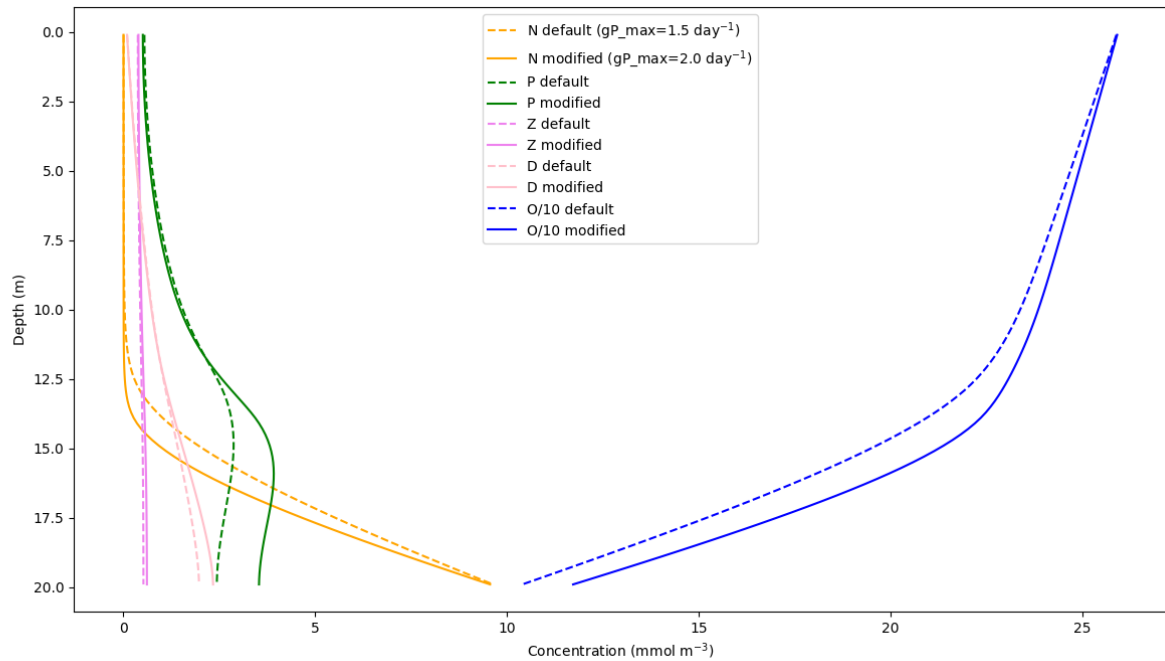


Figure 10: Final NPZDO profiles: phytoplankton maximum growth rate impact

Increasing phytoplankton maximum growth rate has the exact opposite effect compared to increasing light half saturation coefficient. Indeed, phytoplankton concentration increases as well as zooplankton, detritus and oxygen. Phytoplankton concentration bump is deeper than with a lower value of $g_{P,max}$. Furthermore, nutrient concentration decreases.

9 Seasonal succession

To show seasonal succession, we modeled seasonal diffusivity and seasonal light variation as described in sections 2.3 and 2.4. In this way, light follows a sinusoidal variation to simulate annual cycles, and the diffusivity profile follows a hyperbolic tangent transition from surface to deep-water mixing to represent a seasonally varying mixed-layer depth.

The NPZDO model was run over two years, plotting the evolution of the depth distribution of each state-variable concentration over time. The seasonal cycle is primarily driven by light availability and vertical mixing, which in turn regulate nutrient dynamics and the biological response.

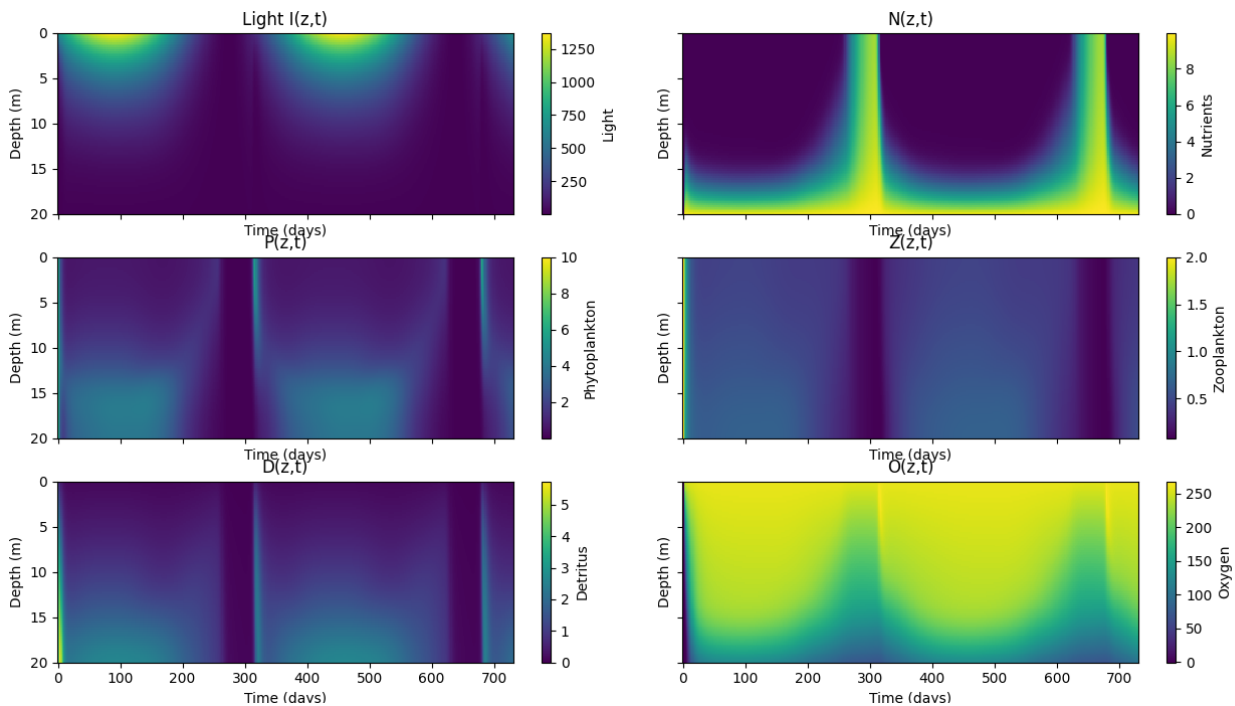


Figure 11: Convergence plot of the NPZDO model after two years with seasonal variation

The NPZDO water column model exhibits a clear seasonal cycle driven primarily by variations in light availability and vertical mixing. During winter, low irradiance and deep mixing maintain high nutrient concentrations throughout the water column while phytoplankton biomass remains low due to light limitation. As light increases in early spring and stratification begins to develop, phytoplankton rapidly utilize the accumulated nutrients, leading to a pronounced surface bloom. Zooplankton respond to this bloom with a temporal lag, increasing their biomass and exerting grazing pressure that eventually reduces phytoplankton concentrations. During summer, strong stratification limits the upward transport of nutrients from deeper waters, resulting in nutrient depletion in the surface layer and relatively low phytoplankton productivity at the surface despite high light availability. Detritus accumulates through mortality and grazing processes and gradually sinks through the water column where it is remineralized, contributing to nutrient regeneration at depth. In autumn, in-

creased mixing replenishes nutrients in the upper layer and triggers a small phytoplankton increase. Oxygen concentrations generally reflect biological activity and mixing processes, with higher values in surface waters due to photosynthesis and lower values at depth where remineralization dominates.

NB: To include seasonal succession, we plotted the solutions using phytoplankton self-shading in light profile, the seasonal light and the seasonal diffusivity (`use_phyto_light_damping = True`, `use_seasonal_light = True`, `use_seasonal_diffusivity = True`).

10 How to improve this model for more accurate representation?

In order to get a more accurate representation, I would add temperature and salinity as state variables. Additionally, we could model the influence of sediments. Moreover, I don't really know how to but we could model human activity impact by determining how close to the coast the water column we are modeling is.

I think priority should be given to the influence of sediments, then to the impact of salinity and temperature. Finally, we could eventually add human activity.

11 Link to my code on GitHub

You can find my code on GitHub at https://github.com/disco00o/25314-NPZD0_Model.